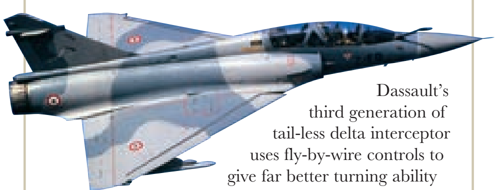
FIGHTING AIRCRAFT

Modern fighters are among the wonders of modern technology. These astonishing machines can typically reach speeds well in excess of Mach 2; climb around 30,000ft (10,000m) a minute to an operational altitude of about 10 miles (16km); execute high-speed jinks and twirls in close combat; and, of course, operate by day and night in all weathers. Reliable engines, improved design, and fly-by-wire controls have made these aircraft much safer to fly than earlier high-performance jets. Immense engine power – up to 50,000lb (22,700kg) total thrust – means that they have been able to grow heavier without losing out on performance. Fitted with an array of radar and infrared devices able to identify enemy aircraft at distance and warn against incoming missiles, their own missile systems are able to engage targets well beyond visual range. They are also very expensive. Economics dictate that even the most dedicated air-superiority fighters end up being used in the ground-attack role too.

STEALTH FIGHTER

The F-117 was totally distinct from other "F"-designated modern aircraft. This subsonic night flier depended on its stealth features for survival in hostile airspace. *See pages 324–25.*

Dassault Mirage 2000C



Dassault's third generation of tail-less delta interceptor uses fly-by-wire controls to give far better turning ability than previously available with this wing form. The aircraft entered French Air Force service in 1988, and 14 were deployed to Saudi Arabia in 1991 as part of Operation Desert Storm, although they did not see combat. There are also a two-seat nuclear and conventional attack versions, the 2000N and 2000D, and the updated multirole Mirage 2000-5.

Engine 21,385lb (9,715kg) thrust SNECMA M53-P2 turbofan

Wingspan 29ft 11in (9.1m) **Length** 47ft 1in (14.3m)

Top speed 1,452mph (2,336kph) (Mach 2.2) **Crew** 1

Armament 2 x 30mm DEFA cannon, 4 x Matra air-to-air missiles; up to 13,890 lb (6,300 kg) attack weapons

Eurofighter (EFA) Typhoon

Studies for a replacement for the Panavia Tornado began in the late 1970s. Development, by a four-nation consortium of Britain, Germany, Italy, and Spain, was long and slow, but the first Eurofighters entered service in 2003. It is a "swing role" aircraft, able to switch from air-to-air to air-to-ground mission configuration in flight.

Engines 2 x 20,000lb (9,086kg) thrust Eurojet EJ 200 turbofan

Wingspan 35ft 11in (11m) **Length** 52ft 4in (16m)

Top speed c.1,320mph (c.2,124kph) (Mach 2) **Crew** 1

Armament 1 x 27mm IWKA-Mauser cannon; 13 weapon carriage points under fuselage and wings

General Dynamics F-111E

The F-111 was the world's first swing-wing aircraft, entering service with the USAF in 1967. Variable geometry enables the aircraft to take off and land with straight wings, but to fly at supersonic speeds with swept wings. The F-111 was also first with automatic terrain-following radar, making it a powerful low-level strike aircraft. Used in Vietnam in 1972–73, against Libya in 1986, and then in Operation Desert Storm in 1991. F-111s were finally retired in 1993.

Engines 2 x 18,500lb (8,392kg) thrust P&W TF30-P-3 turbofan

Wingspan 63ft (19.2m) **Length** 73ft 6in (22.4 m)

Top speed 1,650mph (2,655kph) (Mach 2.5) **Crew** 2

Armament 2 x 750lb (340kg) nuclear/conventional bombs; or 1 x 20mm cannon and 25,000 lb (11,340 kg) of bombs or missiles

Grumman F-14A Tomcat

The Tomcat served as the US Navy's fleet defense fighter from its introduction in 1974 to 2006. It was the first carrier-based aircraft to feature variable-geometry wings. In its day the Tomcat was widely regarded as unsurpassed as a dogfighting aircraft, in large measure due to automatic wing-sweep variation, which adjusted according to the maneuver being undertaken.

Engines 2 x 20,900lb (9,480kg) thrust Pratt & Whitney turbofan

Wingspan 64ft 1in (19.5m) **Length** 62ft 8in (19.1m)

Top speed 1,564mph (2,517kph) (Mach 2.34) **Crew** 2

Armament 1 x 20mm Vulcan rotary cannon; 6 x air-to-air missiles, or up to 14,500lb (6,577kg) attack weapons

General Dynamics F-16 Fighting Falcon

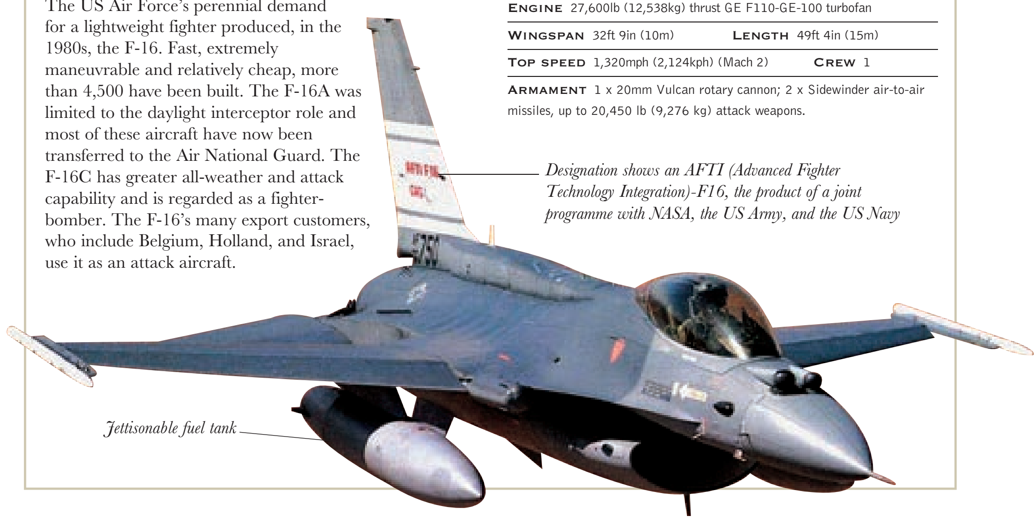
The US Air Force's perennial demand for a lightweight fighter produced, in the 1980s, the F-16. Fast, extremely maneuvrable and relatively cheap, more than 4,500 have been built. The F-16A was limited to the daylight interceptor role and most of these aircraft have now been transferred to the Air National Guard. The F-16C has greater all-weather and attack capability and is regarded as a fighter-bomber. The F-16's many export customers, who include Belgium, Holland, and Israel, use it as an attack aircraft.

Engine 27,600lb (12,538kg) thrust GE F110-GE-100 turbofan

Wingspan 32ft 9in (10m) **Length** 49ft 4in (15m)

Top speed 1,320mph (2,124kph) (Mach 2) **Crew** 1

Armament 1 x 20mm Vulcan rotary cannon; 2 x Sidewinder air-to-air missiles, up to 20,450 lb (9,276 kg) attack weapons.



Designation shows an AFTI (Advanced Fighter Technology Integration)-F16, the product of a joint programme with NASA, the US Army, and the US Navy

Jettisonable fuel tank